

Chaichanok (Man) Vilailuck

Data Science Graduate Student
Bangkok, Thailand

ABOUT

I am a Thai engineer and data scientist in training who is preparing to become the third-generation leader of SAMART, a major Thai telecommunications and ICT group. My long-term goal is to lead Thailand into an AI-enabled era where digital infrastructure, public services, and national communication systems are designed using rigorous data, algorithms, and an understanding of social impact, not just short-term commercial decisions.

My academic path has been intentionally interdisciplinary across industrial engineering, operations research, and data science, so I can operate at the intersection of algorithms, infrastructure, and society.

CONTACTS

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SKILLS

- Python: NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch
- R: Tidyverse, ggplot2
- SQL: Intermediate querying, data wrangling, exploratory analysis
- Prototyping & Tools: Streamlit, Jupyter, Excel
- Supervised learning: logistic regression, decision trees, random forests, gradient boosting
- Unsupervised learning: clustering, PCA, anomaly detection
- Computer Vision: OCR pipelines, image preprocessing, document parsing
- Retrieval-Augmented Generation (RAG), vector databases
- Prompt engineering, model evaluation, system integration
- Model performance analysis, metrics design, error diagnosis
- Stochastic modeling and simulation
- Bayesian statistics
- Statistical inference: hypothesis testing, confidence intervals
- Experimental design and basic A/B testing
- Research & Technical Communication
- Research design and methodological planning

EDUCATION

University of California, Berkeley

M.S., Information and Data Science

GPA: 3.83

Relevant Coursework: Introduction to Data Science Programming, Research Design and Applications, Statistics for Data Science, Fundamentals of Data Engineering, Applied Machine Learning, Data Visualization, Behind the Data: Humans and Values, and Fundamentals of Generative AI

Berkeley, California

August 2024 - (Expected) May 2026

Cornell University (Cornell Tech)

M.Eng., Operations Research and Information Engineering
2025

GPA: 3.97 (Magna Cum Laude Honors)

Relevant Coursework: Optimization Methods, Modeling Under Uncertainty, Applied Data Science & Decision-Making, Applied Machine Learning, Productizing Machine Learning, Product Studio, Trust and Safety, and BigCo Studio

New York, NY

August 2024 - May

University of Illinois Urbana-Champaign

B.S., Industrial Engineering

GPA: 3.56 (Honors)

Relevant Coursework: Deterministic Models in Optimization, Stochastic Processes & Applications, Simulation Modeling for IE, Design & Analysis of Experiments, Linear Algebra with Computational Applications, Production Planning & Control, Human Factors, and Algorithms & Computing

Champaign, IL

August 2020 - May 2024

RESEARCH INTERESTS

1. Robust and Retrieval-Augmented ML Systems

Developing scalable and uncertainty-aware LLM and RAG systems, with a focus on improving retrieval quality, latency, and robustness under noisy or incomplete data. Interested in end-to-end evaluation, adaptive retrieval strategies, and failure-mode analysis for mission-critical applications.

2. Optimization & Stochastic Modeling for Critical Infrastructure

Applying optimization, simulation, and stochastic processes to understand and improve large-scale infrastructure such as communication networks, mobility systems, and digital public services. Interested in robust decision-making under uncertainty and resource-constrained environments.

3. Trustworthy Algorithmic and Sociotechnical Systems

Investigating how algorithmic systems shape governance, incentives, and equity within institutions. Focused on fairness, accountability, and transparency, particularly in regulated or high-stakes settings like telecom, emergency response, and public-sector operations.

4. Human-AI Interaction in High-Stakes Workflows

Designing interfaces, oversight mechanisms, and feedback loops that support non-expert operators using AI systems. Interested in human trust calibration, hybrid decision-making, and the operational safety of AI-assisted workflows across enterprise and public-service environments.

5. Causal Inference for Platforms, Policy, and Behavior

Using causal modeling and experimentation to understand how algorithmic design choices, incentives, or policy interventions influence user behavior and system performance. Interested in applying these methods to platform governance and large-scale institutional decisions.

6. AI Safety, Reliability, and Infrastructure-Aware ML

Studying how ML systems fail in real deployments and developing techniques for robustness, uncertainty quantification, and error mitigation. Interested in designing models that remain reliable under constraints such as bandwidth limits, hardware variability, and noisy data streams.

PROJECTS AND RESEARCH EXPERIENCE

Social Media Warning Labels & Data Science Neutrality Analysis: Societal Computing, Tech Policy, Mixed-Methods Research.

Collaborated on a research paper analyzing whether social media platforms are “neutral carriers of speech” or engineered, data-driven behavioral products. Conducted a legal–technical review of state-level warning-label laws (CO, MN, CA) and evaluated algorithmic engagement mechanisms that undermine claims of speech neutrality. Synthesized research from behavioral psychology, recommender systems, and First Amendment case law to argue that social media interfaces function as designed products with measurable mental health externalities. Contributed to broader debates on AI ethics, platform accountability, and regulatory design.

AI Privacy & Sociotechnical Risk Assessment for Presentation Companion (Capstone Audit): Privacy Frameworks, Sociotechnical Systems, Responsible AI.

Conducted a comprehensive audit of an AI-powered presentation assistant using speech recognition (WhisperX) and RAG-based Q&A modules. Applied Solove’s Taxonomy, Nissenbaum’s Contextual Integrity, and Mulligan et al.’s Sociotechnical Systems analytic to evaluate privacy risks related to continuous audio capture, data processing, and model hallucination. Analyzed system behavior around over-collection, data minimization, consent, and fairness in ASR performance for accented and impaired speech. Produced a structured set of mitigation recommendations including human-in-the-loop verification, contextual disclosures, bias testing, and model card updates aligned with FIPPs and Belmont ethical principles. Demonstrated how design, policy, and algorithmic constraints interact in real AI deployments.

BART Ridership Recovery Study: Experimental Design, Causal Inference, Urban Systems Research.

Designed a mixed-methods experimental study to investigate why Bay Area Rapid Transit ridership has not returned to pre-pandemic levels. Developed a controlled–treatment survey experiment with 20,000 participants (digital and mail distribution) to compare organic vs prompted concerns (safety, cost, cleanliness). Utilized topic modeling, Barnard’s test, Bonferroni corrections, and demographic-stratified analysis to identify statistically significant differences among current, former, and non-riders. Integrated external datasets including BART ridership logs, MTA crime data, and remote-work census data. Proposed policy and operational interventions to restore transit adoption.

Scientific Verifier for Medical Misinformation (BlueSky Social): Trustworthy AI, Information Credibility, Sociotechnical Systems (<https://bsky-scientific-verifier.vercel.app/>).

Created a prototype system to assess the credibility of medical claims online by integrating scientific metadata (h-index, citation networks, publication history). Developed algorithms to contextualize author expertise and used a structured scoring framework to detect high-risk misinformation. Demonstrated how credibility signals can complement LLM-based fact-checking tools.

WORK EXPERIENCE

IBM

Data and Analytics Intern

Bangkok, Thailand
September 2025 - December 2025

- **LLM Retrieval-Augmented Generation (RAG) Optimization:** Designed and evaluated enterprise-grade RAG pipelines using vector databases and large language models. Improved response latency by 0.5–1.0 seconds per query and increased output accuracy by approximately 60% through ablation studies, retrieval strategy experimentation, and error analysis.
- **Developed prompt-engineering frameworks** and response-compression techniques that shortened LLM outputs by ~50%, significantly improving end-user efficiency.
- **Performed robustness testing under noisy retrieval, ambiguous queries, and imperfect knowledge bases**, providing reliability insights to engineering and product teams.

Samart Telcoms PCL

Engineering Consultant Intern

Bangkok, Thailand
June 2025 - August 2025

- **OCR & Computer Vision Pipeline Development:** Built production-level document understanding pipelines to extract structured financial fields from heterogeneous invoice formats. Implemented preprocessing, noise reduction, and rule-based validation logic, reducing manual finance workload by ~30% across partner organizations.
- **Conducted accuracy benchmarking** across multiple vendor templates and error cases, improving model reliability in real deployment conditions.
- **Delivered and configured 200+ secure workstations** for the Provincial Police Telecommunications (PPT) unit, integrating devices into national ICT infrastructure for a project valued at 9M THB.

SLAC National Accelerator Laboratory (Stanford University)

Software Engineering Intern

Menlo Park, CA, USA
June 2023 - August 2023

- **Developed a Python-based automated logging and monitoring system** for grid-integration experiments, version-controlled with Git.
- **Managed AWS S3, Lambda, and IoT Core pipelines** to support secure and scalable research workflows.
- Collaborated in agile sprints, contributing to three successful development cycles for scientific data systems.